Can a carbon-aware VCG rule allocate shared GPU-hours better than first come, first served?

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SUSTAINABLE DEVELOPMENT GOALS

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KEY MESSAGE

A published, carbon-aware VCG rule beats or ties all 720 FCFS arrival orders on value net of carbon (19.2 vs. 16.96 mean), cuts emissions per unit of value (0.72 vs. 0.81 kg), and keeps truthful reporting optimal under C1 + C2 (own carbon charge, linear credit cost), but serves fewer teams (4 vs. 4.93).

01 Question & model

Setting & verified gap

Setting: university GPU queues ignore both project value and carbon.

Gap: FAIR-Compute [5] shows value-blind rules lose about a third of value but leaves carbon and truthful mechanisms open. Truthful cloud auctions price server costs [6]; carbon-aware schedulers shift load without prices [7]. We build the missing rule at an exactly solvable scale.

Strategic model

Players: six teams A–F; each privately knows $v_i \in \{3, 6, 9\}$ and sends one sealed report r_i (static game, incomplete information).

Public: request d_i, emissions $e_i = 0.24\, d_i$ kg CO_2e, capacity $K = 100$, $\lambda = 0.5$.

Rule: pick the feasible group S^* maximizing $\Sigma(r_i - \lambda e_i)$ over all 64 groups; each winner pays the score the others lose (VCG) in credits.

Payoff: $u_i = v_i - \lambda e_i - p_i$ if selected, else 0.

Solution: truthful $r_i = v_i$ is dominant (DSIC) [1–4] under C1 each winner pays its own carbon charge λe_i in credits, and C2 a charged credit has a linear opportunity cost (stated assumption). C3 checks feasibility: a fixed 100-credit budget per round, no top-ups; the largest spend, 60, fits in the tested cases.

Named contributions

Mustafa Ayub Khan: framing and closest work; credit design C2 from his self-challenge; Practical Impacts and SDG targets; checks names in paper and poster.

Temur Akhtamjonov: model and exact solver; FCFS–VCG comparison; GPU-type and measured-emission study; verifies revisions across artifacts.

Gihun Lee: Space and notebook; λ sweep, two-pool and C3 budget checks; Figure 1 annotations, roadmap, two-page paper, Appendix D.

Joint: question, model, and every artifact, reviewed by all three.

Three lenses

Game theory: truthful reporting is each team's best strategy, whatever others report.

Social choice: value net of carbon under capacity; the cost is fewer teams served (4 vs. 4.93).

Mechanism design: a published score rule with VCG prices replaces the queue; λ moves prices, not the winners or the best strategy.

02 Design & evidence

Q1 Economics
How should scarce GPU-hours be allocated for value net of carbon when teams can misreport their private value?

Q2 Computation
Can an exactly solved multi-team VCG beat first come, first served and stay auditable?

Q3 Behavior
Do people report truthfully and understand the rule?



Block width = public request d_i. Shade = private value v_i (dark 9, medium 6, light 3). Emissions $e_i = 0.24\, d_i$ kg CO_2e; $\lambda = 0.5$.

Figure 1. Research questions and setting. Six teams request 120 GPU-hours from a pool of 100. Values are private; requests and emissions are public. Q1 and Q2 are answered by the model and its exact computation, Q3 by play in the Hugging Face Space.

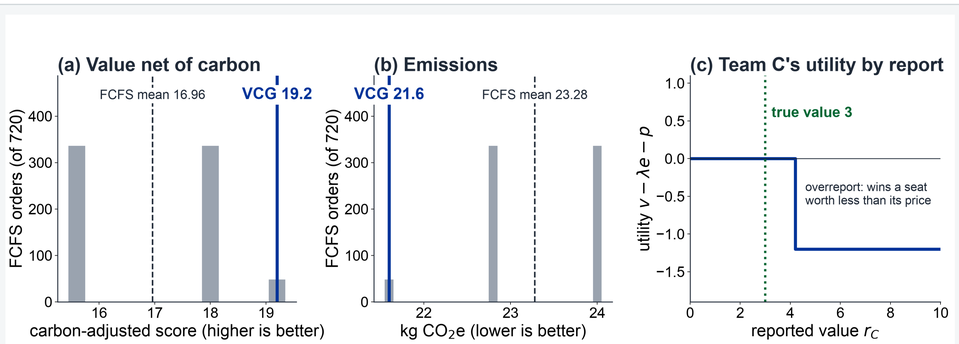


Figure 2. FCFS outcomes across all 720 arrival orders vs. VCG (a, b), and Team C's utility for every report (c).

Result: VCG picks A, B, D, E (value 30, 21.6 kg CO_2e); no FCFS order beats it, but FCFS serves more teams on average.

Parameter change: for λ from 0.1 to 1 the same group wins and truth stays optimal; with two GPU pools emissions fall from 21.6 to 16.6 kg. Winners spend 48–54 of 100 credits.

Verification: run_simulation.py reproduces the outputs; 8 unit tests pass; notebook 10/10 checks PASS; tested commit 2045024.

03 Impact & dialogue

Practical impact & SDG targets

9.5 research capability: value net of carbon 16.96 → 19.2 (simulation only).

9.4 resource efficiency: 0.81 → 0.72 kg CO_2e per value; 0.55 with two pools (simulation only).

13.3 institutional capacity: published λ, fixed credit supply, re-runnable audit log (intended; no adoption).

Stakeholders: research teams, cluster admins, the institution, the public.

Risk → safeguard: heavy λ trades 9.5 for 9.4 → publish λ in advance from local carbon-price data (1.25 is only this instance's threshold); repeated exclusion → history-based credits (planned); credit inflation → fixed 100-credit budget, no top-ups (C3).

Behavioral evidence: five self-plays (exploratory)

Measure	Benchmark	Observed
Truthful reports (r = v)	5 / 5	2 / 5
Overreports	0	3
Selection predicted correctly	—	3 / 5
Plays where a misreport raised utility	0	0

Agreement: no misreport paid off; one overreport bought a seat for 36 credits (u = −3.6).

Gap: three of five overreported (wanting to win; misunderstanding). Reflections were blank; new plays will ask for a reason first.

Feedback, recognition & roadmap

Duet & roundtable (Z. Wang, Q. (Aaron) Wang, FP8): credits are not money → C1 + C2 stated, C3 checked; more GPU types → truth holds with 1 and 2 pools, more types next.

Instructor (Oct 14): conditions beside the claim; two-page paper; SDG targets with evidence status.

Award: Interdisciplinary Mechanism Design Award from FP8.

Roadmap: Econ: credit value over 10 rounds, announcing λ [9]. CS: GPU types with measured emissions, history-based credits [8]. Behavior: reason-before-result study, n ≥ 30.

Acknowledgments

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[3] Groves (1973). Econometrica 41(4), 617–631. doi:10.2307/1914085
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[7] Radovanović et al. (2021). Carbon-aware computing for datacenters. arXiv:2106.11750
[8] Vuppalapati et al. (2023). Karma. USENIX OSDI, 645–662
[9] Royal Swedish Academy of Sciences (2026). Prize in Economic Sciences 2026 (Hall, Woodford)

OPEN MATERIALS

Hugging Face Space (left QR, commit dddebc7):
https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2_Team_A
GitHub (right QR, tested commit 2045024):
https://github.com/dku-comsci-econ206-Autumn2026/FP05-PS2-Team-A
Colab notebook:
https://colab.research.google.com/drive/1xYtctTPKXDszHMidOS7IVz-OByt4edID